



CSCE 633-700: Machine Learning

Linear regression

Overview

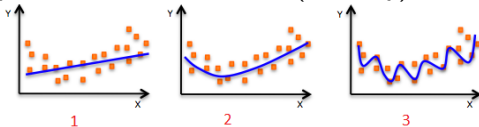
Linear Regression

- Example
- Representation, Evaluation
- Optimization: Closed form solution via Ordinary Least Squares
- Optimization: Numerical solution via Gradient Descent
 - General gradient descent
 - Gradient descent for linear regression (batch, stochastic, mini-batch)
- Non-linear basis function for regression & Overfitting

[Parts of these slides have been adapted from K. Murphy (Machine Learning: A probabilistic perspective), Dr. Andrew Ng's Machine Learning course at Coursera, and CSCI567 Machine Learning (USC, Drs. Sha & Liu)]

Non-Linear Regression

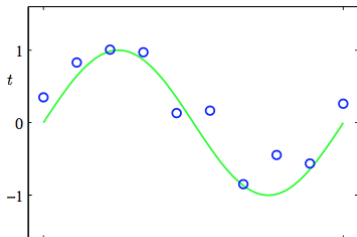
Question: Given a set of training data (red square points), which of the following regression models (blue line) would you choose to fit the data (and why)?



- A) Model 1, since the data depicts an increasing trend
- B) Model 2, since the line best captures the overall trend in the data
- C) Model 3, since the line provides the smallest RSS error

What if the data does not fit a line?

Example: Samples from a sine function



We can use a non-linear basis function

$$\phi(\mathbf{x}) : \mathbf{x} \in \mathbb{R}^D \rightarrow \mathbf{z} \in \mathbb{R}^M$$

We can apply our linear regression model to the new features

$$y_i = \mathbf{w}^T \mathbf{z}_i = \mathbf{w}^T \phi(\mathbf{x}_i)$$

$$RSS(\mathbf{w}) = \sum_{n=1}^N (y_i - \mathbf{w}^T \phi(\mathbf{x}_i))^2, \quad \mathbf{w} \in \mathbb{R}^M$$

$$\text{Example: } \mathbf{x} = [x_1, x_2]^T \in \mathbb{R}^2, \quad \phi(\mathbf{x}) = [x_1, x_2^2, x_1^3 + x_2]^T \in \mathbb{R}^3$$

Non-Linear Basis Function

Residual sum of squares

$$RSS(\mathbf{w}) = \sum_{n=1}^N (y_i - \mathbf{w}^T \phi(\mathbf{x}_i))^2 = (\mathbf{y} - \Phi \mathbf{w})^T (\mathbf{y} - \Phi \mathbf{w})$$

Non-linear design matrix

$$\Phi = \begin{bmatrix} \phi(\mathbf{x}_1)^T \\ \phi(\mathbf{x}_2)^T \\ \vdots \\ \phi(\mathbf{x}_N)^T \end{bmatrix} \in \mathbb{R}^{N \times M}$$

LMS solution with the non-linear design matrix

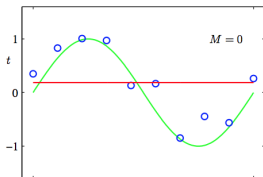
$$\mathbf{w}^{LMS} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y}$$

Non-Linear Basis Function

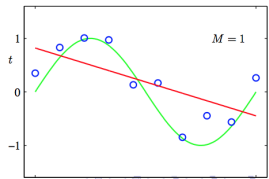
Example: Samples from a sine function

Polynomial basis function $\phi(\mathbf{x}) = [1 \ x \ \dots \ x^M]^T$

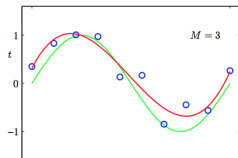
$M = 0$ underfitting



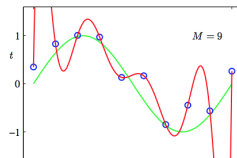
$M = 1$ underfitting



$M = 3$



$M = 9$ overfitting



Overfitting

Weights of high order polynomials are very large

$$y_i = \mathbf{w}^T \mathbf{z}_i = \mathbf{w}^T \phi(\mathbf{x}_i), \quad \mathbf{z}_i = \phi(\mathbf{x}_i) \in \mathbb{R}^M$$

	$M = 0$	$M = 1$	$M = 3$	$M = 9$
w_0	0.19	0.82	0.31	0.35
w_1		-1.27	7.99	232.37
w_2			-25.43	-5321.83
w_3			17.37	48568.31
w_4				-231639.30
w_5				640042.26
w_6				-1061800.52
w_7				1042400.18
w_8				-557682.99
w_9				125201.43